States of Matter Unit Overview KS3

About This Unit

The first part of this unit aims to give pupils an understanding of the particulate nature of matter, the difference in arrangements of particles in solids, liquids and gases based on the particle model, how matter can change from one state to another and the movement of particles in terms of diffusion.

The second half of this unit focuses on mixtures, solubility and how mixtures can be separated using a variety of techniques including filtration, evaporation, distillation and chromatography.

Lessons

1. Solids, Liquids and Gases
2. The Particle Model
3. Changing States
4. Diffusion
5. Mixtures
6. Solubility
7. Filtration
8. Evaporation
9. Distillation
10. Chromatography
11. States of Matter Revision Pack
12. Separation Techniques Revision Pack

Lesson 1: Solids, Liquids and Gases

Included in this Pack

Lesson Presentation

Teaching Ideas

Solids, Liquids and Gases Activity Sheet

Solids, Liquids and Gases Quick Assessment Sheet

Learning Objectives

To understand that matter can exist in three states.

Topics Covered

To define matter.

To state the three types of matter.

To compare and contrast the three states of matter.

Main Pupil Activities

Once pupils have discussed and have shown the key points on 'What is matter?' and 'How do we know it exists,' pupils then work in pairs to identify examples of the three states of matter they can see in the classroom as directed on the PowerPoint.

Pupils then work in small groups to describe the properties of solids, liquids and gases, guided by questions on the PowerPoint.

After the properties key points have been outlined on the PowerPoint, pupils should be encouraged to complete the Solids, Liquids and Gases Activity Sheet, followed by the Solids, Liquids and Gases Quick Assessment Sheet.

Pupils create a Haiku poem about the key points of the three states of matter as a plenary.

Differentiation

By outcome in the starter, the analogy activity, Diffusion Activity Sheet, Diffusion Mini Assessment Sheet, the plenary and the suggested home learning – The Particulate Nature of Matter Homework Sheet.

By task in the analogy activity (support for lower ability pupils and an extension task for higher ability pupils given on the PowerPoint).

National Curriculum Aims

Physics:

- To compare the similarities and differences, including density differences, between solids, liquids and gases.
- To describe the differences in arrangements, in motion and in closeness of particles and link this to explaining changes of state, shape and density.

Resources/Practical Equipment

Lesson 2: The Particle Model

Included in this Pack

Lesson Presentation

Teaching Ideas

The Particle Model Activity Sheet (differentiated for higher and lower ability)

The Particle Model Quick Assessment Sheet

Learning Objectives

To understand how particles are arranged in the three states of matter.

Topics Covered

To identify the arrangement of particles in each state of matter.

To define density.

To link the arrangement of particles in each state of matter to their properties.

Main Pupil Activities

Once pupils have identified the states of matter within the beach picture on the PowerPoint and have recapped the previous lesson, pupils follow the instructions on the PowerPoint and work in pairs to place their stationery into how they think the particles are arranged in solids, liquids or gases.

After outlining the history of the particle model and density key points on the PowerPoint, pupils are encouraged to write a sentence in their book to compare each picture to a state of matter. Extension tasks are given to allow pupils to create their own comparisons to everyday scenarios, and secondly to include the word 'density' in their analogy.

Once key points on the particle model and gas pressure have been discussed on the PowerPoint, pupils should be encouraged to complete one of The Particle Model Differentiated Activity Sheets, followed by The Particle Model Quick Assessment Sheet.

Pupils follow the instructions on the PowerPoint to complete a '3, 2, 1 Lift Off' plenary. Pupils write in their books three keywords, two facts they have learned today and one question about the topic they have.

Differentiation

By outcome in the starter activity, arranging stationery activity, the analogy activity, The Particle Model Differentiated Activity Sheet, The Particle Model Mini Assessment Activity Sheet and the plenary.

By task in The Particle Model Differentiated Activity Sheet (higher and lower ability activity sheets available).

National Curriculum Aims

Physics:

- To compare the similarities and differences, including density differences, between solids, liquids and gases.
- To describe the differences in arrangements, in motion and in closeness of particles and link this to explaining changes of state, shape and density.
- To identify atoms and molecules as particles.

Working Scientifically:

- To understand that scientific methods and theories develop as earlier explanations are modified to take account of new evidence and ideas, together with the importance of publishing results and peer review.

Resources/Practical Equipment

Lesson Pack

Optional homework: Particle Model Homework Activity Sheet.



Lesson 3: Changing States

Included in this Pack

Changing States PowerPoint

Teaching Ideas

Stearic Acid Practical Activity Sheet

Changing States Quick Assessment Sheet

Learning Objectives

To understand how matter can change from one state to another.

Topics Covered

To list the different changes of state.

To identify which change of state is happening.

To describe what happens as matter changes between states.

Main Pupil Activities

Once pupils have discussed what connects the pictures on the PowerPoint and have recapped the previous lesson, pupils follow the instructions on the PowerPoint to identify keywords for changing states.

Pupils should be encouraged to complete the gap fill definitions on the PowerPoint, before being given a brief introduction to the stearic acid practical in order to complete the Stearic Acid Practical Activity Sheet.

Pupils should then be encouraged to complete the five quick questions on the Stearic Acid Practical Activity Sheet, followed by the Particle Model Quick Assessment Sheet. Or they could plot their data as a graph using the prompts on the Stearic Acid Practical Activity Sheet.

Differentiation

By outcome in the starter activity, identifying keywords for changing states activity, Stearic Acid Practical Activity Sheet, Changing States Quick Assessment Sheet and the plenary activity.

By support in the gap fill definition activity. There are letter prompts in the gap fill on the PowerPoint and on the Stearic Acid Practical Activity Sheet. (A pre-drawn results table is also available.)

By task in the gap fill definition activity (extension question on the PowerPoint).

National Curriculum Aims

Physics:

- To describe the differences in arrangements, in motion and in closeness of particles and link this to explaining changes of state, shape and density.
- To describe the difference between chemical and physical changes.
- To explain the changes with temperature in motion and spacing of particles.

Working Scientifically:

- To make and record observations and measurements using a range of methods for different investigations; and evaluate the reliability of methods and suggest possible improvements.

Resources/Practical Equipment

Water bath set to approximately 70°C (before the lesson commences).

Per pair/group of pupils:

Stearic acid in a boiling tube with a thermometer placed inside (placed in the water bath before the lesson commences).

stopwatch

Health and Safety

Schools should prepare their own risk assessments for the stearic acid practical. Common procedures include ensuring pupils are stood up throughout the practical, pupils and staff are wearing goggles, long hair is tied back and no food or drink is consumed. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. Spilt water should be wiped up immediately to prevent slips and falls. The water bath should have a 'caution – hot water' sign attached.

Pupils are reminded of some of these typical rules on the PowerPoint and also on the Stearic Acid Practical Activity Sheet.

Lesson 4: Diffusion

Included in this Pack	Learning Objectives	Topics Covered	Main Pupil Activities
Diffusion PowerPoint Teaching Ideas Diffusion Investigation Activity Sheet Diffusion Quick Assessment Sheet Key Revision Facts Particulate Nature of Matter Homework Sheet	To understand how particles can move from one area to another.	To define diffusion. To explain how diffusion works in terms of the particle model. To predict the movement of particles based on factors.	Once pupils have discussed how the smell of burnt toast moves around the house and doesn't last forever as prompted on the PowerPoint, pupils are introduced to the definition of diffusion and should be encouraged to note this down whilst considering which of the three states of matter could diffuse. Once pupils have watched a demonstration of potassium permanganate in water, pupils should be encouraged to complete the analogy activity on the PowerPoint. Pupils should be allowed access to the five scented balloons in order to complete the first task on the Diffusion Investigation Activity Sheet before continuing onto the rest of the activity sheet. Once the four factors affecting diffusion and the mechanisms behind these are explained, alongside Brownian motion, pupils should be encouraged to complete the Diffusion Quick Assessment Sheet. Following this, pupils should work in pairs to identify the errors in a pretend student's conversation about diffusion on the PowerPoint as a plenary activity.

Differentiation	National Curriculum Aims	Resources/Practical Equipment	Health and Safety
By outcome in the starter activity, analogy activity, Diffusion Investigation Activity Sheet and the plenary activity. By support on the PowerPoint to assist with the synoptic style question 5 on Diffusion Investigation Activity Sheet. Also, the Key Revision Facts Sheet to assist with the suggested homework activity Particulate Nature of Matter Homework Sheet. By task in the definition activity (extension tasks of adding in extra terminology to the definition) and the analogy activity (extension tasks of adding keywords to answers and creating own analogies).	Chemistry: • To define diffusion in terms of the particle model. Physics: • To understand that diffusion in liquids and gases is driven by differences in concentration. • To understand Brownian motion in gases. Working Scientifically: • To make predictions using scientific knowledge and understanding.	A large glass beaker filled with hot water. A large glass beaker filled with cold water. A small amount of potassium permanganate. spatula Safety goggles for pupils and staff. Five different coloured balloons filled with various smells (e.g. coffee, vinegar, milkshake powder etc.) and a personalised teacher answer sheet.	Schools should prepare their own risk assessments for potassium permanganate demonstration. Common procedures include ensuring pupils are stood up throughout the demonstration, pupils and staff are wearing goggles, long hair is tied back and no food or drink is consumed. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. Spilt water should be wiped up immediately to prevent slips and falls. The glass beaker of hot water should have a 'caution - hot water' sign attached. Schools should prepare their own risk assessments for the scented balloons activity, ensuring staff and students are aware of any food allergies and even rubber/latex allergies.

Lesson 5: Mixtures

Included in this Pack	Learning Objectives	Topics Covered	Main Pupil Activities
Lesson Presentation Teaching Ideas Mixtures Activity Sheet Mixtures Quick Assessment Sheet	To understand how mixtures look at a particle level.	To define a mixture and state examples. To compare mixtures to compounds. To model air as a mixture.	Once pupils have discussed whether making a salad is a physical or chemical change and are reminded of the differences between these (on the PowerPoint), pupils observe a teacher led demonstration of iron filings and sulfur powder as separate elements, which are then heated and combined into a compound. Pupils should then be encouraged to complete a definitions cloze style activity and then apply this to four particle pictures as prompted on the PowerPoint. Once pupils have been introduced to the notion of mixtures and justified some examples on the PowerPoint, they should be encouraged to complete the Mixtures Activity Sheet where access to coloured sticky dots or coloured pens is required. After this, pupils should complete the Mixtures Quick Assessment Sheet. As a plenary, pupils should apply today's knowledge and keywords to a case study on a smart memory alloy called nitinol on the PowerPoint.

Differentiation	National Curriculum Aims	Resources/Practical Equipment	Health and Safety
By outcome in the starter, justifying mixtures pictures activity, Mixtures Activity Sheet, Mixtures Quick Assessment Sheet and the plenary activity. By support in the definitions activity (letter prompts on the PowerPoint to support with writing the definition of a mixture for lower ability pupils), identifying mixtures activity (use of a periodic table to help identify elements for lower ability pupils), and justifying mixtures activity (use of notes to justify pictures of mixtures for lower ability pupils). By task in the definition activity (higher ability pupils can create own definition of a compound), in the identifying activity (higher ability pupils can name the elements and compounds present in the pictures), and an extension activity on fractional distillation on the Mixtures Activity Sheet for higher ability pupils.	Chemistry: • To describe the differences between atoms, elements and compounds. • To define a mixture. • To state chemical reactions as the rearrangement of atoms. Physics: • To describe the difference between chemical and physical changes.	Access to a gas tap. safety goggles Bunsen burner heatproof mat Small magnet covered in cling film to stop iron filings from adhering. Weighing boat or a watch glass. Small amount of sulfur powder (approximately 4g). Small amount of iron filings (approximately 7g). Note: only heat 2g of the combination of iron filings and sulfur from the 7:4 ratio above. One heatproof glass test tube. Mineral wool plug for end of test tube. One pair of test tube tongs or boss and clamp and clamp stand. Small circular 'sticky dots' in five different colours (approximately one small sheet per pupil) or five different coloured felt tips.	Schools should prepare their own risk assessments for the iron filings and sulfur powder demonstration. Common procedures include ensuring pupils are stood up throughout the demonstration, pupils and staff are wearing goggles throughout due to the fine nature of the iron filings, long hair is tied back and no food or drink is consumed. The room should be well ventilated and asthmatics should be known to staff and notified beforehand, due to the sulfur vapours produced. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. The Bunsen burner should be positioned in the centre of the desk and tongs should be used to hold the test tube into the flame, with the mineral wool plug pointing away from any person. A damp rag should be nearby to extinguish the sulfur vapours to prevent them from catching fire and placed over the mineral wool plug if needed.

Lesson 6: Solubility

Included in this Pack

Lesson Presentation

Teaching Ideas

Planning an Experiment into Solubility Activity Sheet (differentiated for higher and lower ability)

Solubility Quick Assessment Sheet

Learning Objectives

To plan an investigation on solubility.

Topics Covered

To define keywords in the solubility topic.

To apply solubility keywords to situations.

To design a safe and valid experiment.

To describe how temperature affects solubility.

Main Pupil Activities

Once pupils have discussed three questions surrounding adding sugar to a cup of tea as prompted on the PowerPoint, pupils should be encouraged to draw a cup of tea and annotate the three keywords 'solute', 'solvent' and 'solution' next to each part of the tea. Two additional keywords are also introduced and pupils should be encouraged to note these down.

Pupils then follow one of the differentiated Planning an Experiment into Solubility Activity Sheets to plan a safe and valid experiment into dissolving sugar in water at different temperatures. If time, resources and your school permit, pupils can also perform this experiment during the lesson. This should be followed by the Solubility Quick Assessment Sheet.

As a plenary activity, pupils are shown three pictures and asked to spot as many keywords in the pictures as prompted on the PowerPoint.

Differentiation

By outcome in the starter, keyword activity, Planning an Experiment into Solubility Activity Sheets, Solubility Quick Assessment Sheet and the plenary.

By support in Planning an Experiment into Solubility Activity Sheet (differentiated for higher and lower ability) and also the level of teacher or teaching input to complete the activity sheet. Also if the practical is performed in this lesson, the level of teacher or teaching assistant input in order to safely carry out the practical.

National Curriculum Aims

Chemistry:

- To define a mixture and dissolving.

Working Scientifically:

- To select, plan and carry out the most appropriate types of scientific enquiries to test predictions, including identifying independent, dependent and control variables.
- To use appropriate techniques, apparatus, and materials during fieldwork and laboratory work, paying attention to health and safety.
- To make predictions using scientific knowledge and understanding.

Resources/Practical Equipment

If students finish planning the experiment and there is time to carry it out, they will need the following per pair or small group:

- a small bag of sugar;
- three small 25ml glass beakers;
- one glass rod;
- access to hot water/a kettle;
- one thermometer.

Health and Safety

Schools should prepare their own risk assessments if pupils carry out the solubility practical this lesson. Common procedures include ensuring pupils are stood up throughout the experiment, pupils and staff are wearing goggles, long hair is tied back and no food or drink is consumed. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. Spilt water should be wiped up immediately to prevent slips and falls and especially near electrical sockets on desks. The source of hot water (e.g. kettle) should have a 'caution – hot water' sign attached.

Lesson 7: Filtration

Included in this Pack

Lesson Presentation

Teaching Ideas

Filtration Invention Activity Sheet

Filtration Invention Peer Assessment Grid

Filtration Quick Assessment Sheet

Learning Objectives

To understand how filtration works and its everyday importance.

Topics Covered

To define keywords relating to filtration.

To apply keywords to real-life situations.

To design a filter and communicate ideas clearly.

Main Pupil Activities

Once pupils have discussed what links the five pictures on the PowerPoint and have been given a brief introduction to filtering and shown examples, pupils should be encouraged to note down the four keywords, 'filter, 'filtrate', 'residue' and 'filtration', and their definitions.

After being given background information on dirty water, pupils should then gather into small groups to complete the Filtration Invention Activity Sheet to design a water filter and prepare a sales 'pitch' to present to the class. If time permits during the lesson, all pitches could be presented to the class, with the teacher acting as the 'investor'. There is also a Peer Assessment Grid for pupils to complete upon watching their classmates' presentations and pupils could vote for their favourite pitch.

Afterwards, pupils should complete the Filtration Quick Assessment Sheet.

As a plenary, pupils think of six keywords from today's lesson and write them in order of importance as shown on the diagram, similar to a 'diamond nine' activity.

Differentiation

By outcome in the starter, Filtration Invention Activity Sheet, Filtration Quick Assessment Sheet and the plenary.

By support with the Filtration Invention Activity Sheet – teacher or teaching assistant input to help with filtration invention, alongside pre-written scaffolded sales pitch structure for lower ability pupils.

By task in the keywords activity – extension task of drawing filtering apparatus, and applying keywords to the diagram for higher ability pupils.

National Curriculum Aims

Chemistry:

- To describe simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography.

Working Scientifically:

- To ask questions and develop a line of enquiry based on observations of the real world, alongside prior knowledge and experience.
- To use appropriate techniques, apparatus, and materials during fieldwork and laboratory work, paying attention to health and safety.
- To make predictions using scientific knowledge and understanding.

Resources/Practical Equipment

Lesson 8: Evaporation

Included in this Pack

Lesson Presentation

Teaching Ideas

Separating Salt from Rock Salt Activity Sheet (differentiated for higher and lower ability)

Separating Salt from Rock Salt Practical Teacher Feedback Grid

Evaporation Quick Assessment Sheet

Differentiation

By outcome in the starter, Separating Salt from Rock Salt Activity Sheet, discussing the formation of salt lakes, Evaporation Quick Assessment Sheet and the plenary.

By support on the Separating Salt from Rock Salt Activity Sheet (a method is pre written for lower ability pupils) and also the level of teacher or teaching assistant input to safely carry out the practical for lower ability pupils.

By task on the Separating Salt from Rock Salt Activity Sheet – a task of sequencing a method is given to higher ability pupils. Also, the final question of the plenary where higher ability pupils are encouraged to think why some of the steps for making copper sulfate crystals are different to making pure salt crystals.

Learning Objectives

To understand how to use evaporation as a separation technique.

National Curriculum Aims

Chemistry:

- To describe simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography.

Working Scientifically:

- To evaluate risks.
- To select, plan and carry out the most appropriate types of scientific enquiries to test predictions, including identifying independent, dependent and control variables.
- To use appropriate techniques, apparatus, and materials during fieldwork and laboratory work, paying attention to health and safety.

Topics Covered

To identify scientific apparatus.

To safely follow a method.

To apply keywords to a practical.

Resources/Practical Equipment

Rock Salt Practical – pupils will need the following per pair or small group;

- rock salt;
- pestle and mortar;
- 2 x 250ml glass beakers;
- funnel and filter paper;
- glass rod;
- heatproof mat;
- Bunsen burner;
- tripod;
- evaporating basin.

Optional: Premade copper sulfate crystals in a sealed petri dish for pupils to observe as part of the plenary activity.

Main Pupil Activities

Once pupils have completed the starter literacy task by making as many words out of 'evaporation' as possible in three minutes, pupils are given a recap about mixtures, including the definition, and how they link to separation techniques.

Pupils are reminded of the definition and conditions of evaporation, with pictorial support of the arrangements of particles in a liquid compared to a gas. A brief background on rock salt is presented to the class, and pupils then follow one of the differentiated Separating Salt from Rock Salt Activity Sheets to carry out a safe practical, with the aim of separating pure salt from rock salt. There are questions to answer at the end of the practical.

Pupils are shown pictures and information about salt lakes and halophiles to make synoptic links to biology. Pupils are shown how evaporation can occur on large scales. Pupils should then complete the Evaporation Quick Assessment Sheet.

As a plenary activity, pupils are shown a picture of copper sulfate crystals on the PowerPoint, and if possible, have some ready to observe in a sealed petri dish to pass around the class. Pupils are shown a pictorial method on the PowerPoint for making copper sulfate crystals. In pairs, they should try to spot the similarities and differences between their rock salt practical in today's lesson and the copper sulfate method.

Health and Safety

Schools should prepare their own risk assessments for the rock salt practical. Common procedures include ensuring pupils are stood up throughout the demonstration, pupils and staff are wearing goggles throughout due to the fine nature of the iron filings, long hair is tied back and no food or drink is consumed. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. Pestle and mortars, and evaporating basins should also be handled with care and broken pieces disposed of carefully by staff. The Bunsen burner should be positioned in the centre of the desk, on top of a heatproof mat and only packed away once cooled down.

Schools should prepare their own risk assessments for the copper sulfate crystal observation. Common procedures include ensuring the copper sulfate crystals are in a sealed petri dish.

Lesson 9: Distillation

Included in this Pack

Lesson Presentation

Teaching Ideas

Distillation Apparatus Activity Sheet

Distillation Sequencing Cards (differentiated for higher and lower ability students)

Distillation Information Cards

Distillation Real World Applications Activity Sheet (differentiated for higher and lower ability students)

Distillation Quick Assessment Sheet

Differentiation

By outcome in the starter, Distillation Apparatus Activity Sheet, Distillation Sequencing Cards, Distillation Real World Applications Activity Sheet, Distillation Quick Assessment Sheet and the plenary.

By support in the distillation demonstration (the teacher can vary input and type of questioning), on the Distillation Sequencing (beginning and end steps in the correct order for lower ability students) and on the differentiated Distillation Real World Applications Activity Sheet – there are more scaffolded and accessible questions for lower ability students compared to more challenging questions for higher ability students.

Learning Objectives

To understand how to use distillation as a separation technique.

National Curriculum Aims

Chemistry:

- To describe simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography.

Working Scientifically:

- To make predictions using scientific knowledge and understanding.
- To interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions.

Topics Covered

To identify scientific apparatus.

To explain how distillation works.

To recognise the everyday uses of distillation.

Resources/Practical Equipment

Distillation apparatus needed for demonstration:

- Access to gas tap and water tap (in close proximity to each other).
- Safety goggles for the teacher and all students.
- Small amount of ink mixed with water.
- Bunsen burner
- heatproof mat
- Tripod and gauze (if required)
- Distilling flask/round bottomed flask
- thermometer
- Condenser with rubber tubes attached.
- glass beaker
- Clamp stands and bosses

Main Pupil Activities

Once pupils have described the picture on the PowerPoint using as many keywords from this topic as they can in two minutes, pupils are then given a recap of the two separation techniques covered so far.

Pupils follow the Distillation Apparatus Activity Sheet to apply their knowledge of changing states and scientific apparatus to guess what happens in distillation, guided by the questions on the activity sheet with no teacher input.

Once pupils have watched a teacher led demonstration of simple distillation of ink and water, pupils should be encouraged to complete the differentiated Distillation Sequencing Cards by placing sentences in the correct order to explain the stages in distillation.

To contextualise the lesson, there are three Distillation Real World Applications Information Cards, which could be cut up and placed either around the room, or swapped between pupils. Pupils read an information card on either fractional distillation of crude oil, fractional distillation of air or simple distillation of seawater and complete the matching section on one of the differentiated Distillation Real World Applications Activity Sheet.

Pupils should then complete the Distillation Quick Assessment Sheet.

As a plenary, pupils are prompted to make two lists of 'Apparatus I Knew before Today' and 'Apparatus I Now Know from Today'.

Health and Safety

Schools should prepare their own risk assessments for the simple distillation of ink and water demonstration. Common procedures include ensuring pupils are stood up throughout the demonstration, pupils and staff are wearing goggles, long hair is tied back and no food or drink is consumed. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. The Bunsen burner should be positioned in the centre of the desk and all glassware should be fixed securely in place. The production of hot water, and therefore steam, should be noted by the teacher in case of burns and scalds, and apparatus should be left to cool before packing away.

Lesson 10: Chromatography

Included in this Pack

Lesson Presentation

Teaching Ideas

Solving Crimes Using Chromatography Activity Sheet

Chromatography Quick Assessment Sheet

Learning Objectives

To understand how to use chromatography as a separation technique.

Topics Covered

To create a chromatogram.

To explain how chromatography works.

To recognise the everyday uses of chromatography.

To identify different separation techniques.

Main Pupil Activities

Once pupils have listed colours in their chosen section of the picture on the PowerPoint and identified any primary and secondary colours, pupils are shown a recap about the three separation techniques covered in the topic so far and given a brief introduction to the chromatography, using solubility key words and also given a brief history of technique.

Pupils are then shown a six step sequence of how to perform chromatography on the PowerPoint. This could also be modelled by the teacher simultaneously to show pupils how to set up and create a chromatogram. Pupils are then introduced to the chromatography mystery to solve using the PowerPoint slides and the pre-made chromatogram should be shown to the class as a reference, in order for pupils to then complete the Solving Crimes Using Chromatography Activity Sheet.

To contextualise the lesson, pupils are asked to discuss and then discover how chromatography is used in a variety of different ways and pupils should then be encouraged to complete the Chromatography Quick Assessment Sheet. As a plenary activity, pupils could work in pairs to identify the four separation techniques as shown on the PowerPoint (A to D). Pupils must then decide which technique to use in each situation described, to separate out the mixture. There is also a challenge question asking pupils to explain why they chose that technique to justify their answers.

Differentiation

By outcome in the starter, Solving Crimes Using Chromatography Activity Sheet, Chromatography Quick Assessment Sheet and the plenary.

By support with the Solving Crimes Using Chromatography Activity Sheet (the level of teacher or teaching assistant support required to safely complete the chromatography practical and associated questions) and also in Chromatography Real World Applications on the PowerPoint (the level of teacher input to elicit ideas on real world applications of chromatography).

By task in starter activity (a task of identifying primary and secondary colours is given), in Chromatography Real World Applications on the PowerPoint (if the teacher wishes to generally discuss gas chromatography with higher ability students) and in the plenary on the PowerPoint (students are asked to justify their answers as an extension task).

National Curriculum Aims

Chemistry:

- To describe simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography.

Working Scientifically:

- To pay attention to objectivity and concern for accuracy, precision, repeatability and reproducibility.
- To make and record observations and measurements using a range of methods for different investigations; and evaluate the reliability of methods and suggest possible improvements.
- To interpret observations and data, including identifying patterns and using observations, measurements and data to draw conclusions.

Resources/Practical Equipment

- ruler
- pencil
- three black pens
- A pre-made chromatogram using one of the above pens (water soluble only) to be used as Mr Rye's reference chromatogram.

Each student will need:

- a strip of chromatography or filter paper;
- a small glass beaker;
- a spare pencil or piece of straw and a drawing pin;
- a small amount of water or ethanol to use as the solvent;
- safety goggles.

Health and Safety

Schools should prepare their own risk assessments for the chromatography demonstration and class practical. Common procedures include ensuring pupils and staff are wearing goggles (particularly if ethanol is used), long hair is tied back and no food or drink is consumed. Ensure all bags and coats are safely put away. Glassware should be handled carefully and broken pieces disposed of by staff. Ethanol should be administered by staff in the smallest volume possible. The vapours of ethanol should not be inhaled and the room should be well ventilated. Ethanol is also highly flammable. Methylated spirit (IDA) is also highly flammable, as well as being harmful. Neither ethanol nor methylate spirit should be used near naked flames. Make sure equipment for extinguishing fires is nearby.

Lesson 11: States of Matter Revision Pack

Included in this Pack

States of Matter Student Revision Checklist

Properties of Solids, Liquids and Gases Cut and Stick Activity Sheet

Solids, Liquids and Gases Match and Draw Activity Sheet

Particles and States of Matter Glossary

Solids, Liquids and Gases 'If I Was' PowerPoint

Changes of State Match and Draw Activity Sheet

Changes of State 'Can You Guess' Cards

Changing States PowerPoint Quiz

States of Matter Crossword

States of Matter Assessment Package

Learning Objectives

To consolidate knowledge of the particle model and changes of state.

Topics Covered

To self-assess progress in this topic.

To further strengthen and deepen knowledge of this topic.

To apply knowledge to exam style questions.

Main Pupil Activities

Suggested outline of the lesson:

Starter: Students complete States of Matter Student Revision Checklist or Progress Sheet 1 in the States of Matter Assessment Package in order to identify their areas of strength and areas of improvement, which will be worked on during today's lesson.

Main Activities: Students should choose two or three activities to complete from a mixture of demands (listed in order of increasing demand):

Properties of Solids, Liquids and Gases Cut and Stick Activity Sheet (low demand)

Changes of State Match and Draw Activity Sheet (low demand)

Solids, Liquids and Gases Match and Draw Activity Sheet (low demand)

Particles and States of Matter Glossary (mid demand)

States of Matter Crossword (mid demand)

Changes of State 'Can You Guess' Cards (mid/high demand)

Test Yourself 1, 2 and 3 Activity Sheets in the States of Matter Assessment Package (high demand)

Absent or lower ability students may find the Key Revision Facts sheet in the States of Matter Assessment Package helpful.

These activities should be punctuated by teacher input every 10 to 15 minutes to perform a whole class activity from the following:

Solids, Liquids and Gases 'If I was' PowerPoint

Changing States PowerPoint Quiz

Students should then complete the Exam Style Questions in the States of Matter Assessment Package and self-assess their answers.

Students could take home a copy of Progress Sheet 2 in the States of Matter Assessment Package to help organise their revision.

Differentiation

By outcome in the main activities, whole class activities and the exam style questions.

By support in terms of teacher or teaching assistant input during the main activities and also access to the Key Revision Facts sheet in the States of Matter Assessment Package for lower ability or absent students.

By task in the choice of main activities.

National Curriculum Aims

Chemistry:

- To describe the properties of the different states of matter (solid, liquid and gas) in terms of the particle model, including gas pressure.
- To explain changes of state in terms of the particle model.

Physics:

- To state the conservation of material and of mass, and reversibility, in melting, freezing, evaporation, sublimation, condensation, dissolving.
- To explain the changes with temperature in motion and spacing of particles.

Resources/Practical Equipment

Optional: Mini whiteboards for Changing States PowerPoint Quiz and Solids, Liquids and Gases 'If I was' PowerPoint.

Lesson 12: States of Separating Mixtures Revision Pack

Included in this Pack

Separation Techniques Student Revision Checklist

Solute, Solvent and Solution Match and Draw Activity Sheet

Separating Techniques Match and Draw Activity Sheet

Mixtures Glossary

Pure and Impure Substances 'Can You Guess?' Cards

Separating Mixtures Circular Dominoes

Separation PowerPoint Quiz

Pure and Impure PowerPoint Quiz

Separating Techniques 'If I Were' PowerPoint

Separation Techniques Assessment Package

Learning Objectives

To consolidate knowledge of mixtures, solubility and the four separation techniques.

Topics Covered

To self-assess progress in this topic.

To further strengthen and deepen knowledge of this topic.

To apply knowledge to exam style questions.

Main Pupil Activities

Suggested outline of the lesson:

Starter: Students complete Separation Techniques Student Revision Checklist or Progress Sheet 1 in the Separation Techniques Assessment Package in order to identify their areas of strength and areas of improvement, which will be worked on during today's lesson.

Main Activities: Students should choose two or three activities to complete from a mixture of demands (listed in order of increasing demand):

Solute, Solvent and Solution Match and Draw Activity Sheet (low demand)

Separating Techniques Match and Draw Activity Sheet (low demand)

Mixtures Glossary (mid demand)

Pure and Impure Substances 'Can You Guess?' Cards (mid/high demand)

Separating Mixtures Circular Dominoes (mid/high demand)

Test Yourself 1, 2 and 3 Activity Sheets in the Separation Techniques Assessment Package (high demand)

Absent or lower ability students may find the Key Revision Facts sheet in the Separation Techniques Assessment Package helpful.

These activities should be punctuated by teacher input every 10 to 15 minutes to perform a whole class activity from the following:

Separation PowerPoint Quiz

Pure and Impure PowerPoint Quiz

Separating Techniques 'If I Were' PowerPoint

Students should then complete the Exam Style Questions in the Separating Techniques Assessment Package and self-assess their answers.

Students could take home a copy of Progress Sheet 2 in the Separating Techniques Assessment Package to help organise their revision.

Differentiation

By outcome in the main activities, whole class activities and the exam style questions.

By support in terms of teacher or teaching assistant input during the main activities and also access to the Key Revision Facts sheet in the Separating Techniques Assessment Package for lower ability or absent students.

By task in the choice of main activities.

National Curriculum Aims

Chemistry:

- To describe the differences between atoms, elements and compounds.
- To define a mixture and dissolving.
- To state chemical reactions as the rearrangement of atoms.
- To describe simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography.

Resources/Practical Equipment

Optional: Mini whiteboards for Separation PowerPoint Quiz, Pure and Impure PowerPoint Quiz and Separation Techniques 'If I was' PowerPoint.